A Candidate Certificate-Backed Proof that $z(20) = 6$

The cochromatic number of graphs on twenty vertices (Erdős Problem #758)

Prepared from the local `z20_repro` verification bundle

26 July 2026

Status. This is an **unrefereed candidate computer-assisted proof**, offered for independent mathematical and computational scrutiny. The local certificates have been replayed and the feasible exhaustive computations have been rerun. No human peer review, external reproduction, or end-to-end proof-assistant formalisation is claimed.

Abstract

For a finite simple graph G, let $\zeta(G)$ be the least number of classes in a partition of $V(G)$ such that each class induces either a clique or an independent set, and put

$$z(n) = \max\{\zeta(G) : |V(G)| = n\}.$$

The next value listed as unresolved is $z(20)$, known to be either 6 or 7. This note presents a candidate computer-assisted proof that $z(20) = 6$.

The lower bound is a hand proof using the Paley graph on 17 vertices. For the upper bound, every 20-vertex graph is shown to contain two vertex-disjoint homogeneous 4-sets. If one fixes a first such set, a putative counterexample leaves a homogeneous-4-free graph on 16 vertices. Orderly generation gives exactly two such graphs up to isomorphism. Each leaves 64 undetermined cross edges and yields a 64-variable CNF with 104,524 clauses. Both formulas have certificate-backed refutations. Removing the two homogeneous 4-sets leaves 12 vertices, and a separate exhaustive computation proves $z(12) \leq 4$.

The note separates the mathematical reduction, the exact encoding lemma, the proof-certificate checks, and the remaining trusted-computing assumptions. In particular, the SAT checkers certify the two CNFs, not the graph-theoretic theorem by themselves.

Keywords: cochromatic number; Ramsey graph; exhaustive generation; SAT certificate; DRUP; LRAT.

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1 Statement, scope, and proof architecture

Cocolouring was introduced by Lesniak-Foster and Straight [1]. The finite extremal question considered here is recorded as Erdős Problem #758 [2]. This document is based on the local repository `/Users/admin/z20_repro` and the public artifact snapshot [8].

Definition 1.1. A set $S \subseteq V(G)$ is *homogeneous* if $G[S]$ is complete or edgeless. A *cocolouring* of G is a partition of $V(G)$ into homogeneous classes. Its minimum number of classes is the *cochromatic number* $\zeta(G)$. Finally,

$$z(n) = \max\{\zeta(G) : G \text{ is a finite simple graph on } n \text{ vertices}\}.$$

A $(4,4)$-graph is a graph with neither a K_4 nor an independent 4-set.

Theorem 1.2 (Main result)**.**

$$\boxed{z(20) = 6}.$$

The proof has two unequal halves. The lower bound is entirely hand-checkable. The upper bound depends on three finite computations:

Obligation	Role in the proof
Catalogue	There are exactly two $(4,4)$-graphs on 16 vertices, up to isomorphism; the same run supplies a 17-vertex survivor and no 18-vertex survivor.
Fixed-core SAT	For neither 16-vertex core can the 64 cross edges be chosen so as to avoid two disjoint homogeneous 4-sets.
Residual cocolouring	Every 8-vertex graph is 3-cocolourable, and every $(4,4)$-graph on 12 vertices is 4-cocolourable.

The logical flow is

$$\begin{aligned} P_{17} \text{ has } \alpha(P_{17}), \omega(P_{17}) \le 3 \quad &\Longrightarrow \quad z(20) \ge 6, \\ R(4,4) = 18 + \text{two 16-vertex cores} + \text{two certified UNSAT results} \\ &\Longrightarrow \text{ two disjoint homogeneous 4-sets in every } G_{20} \\ &\Longrightarrow 2 + z(12) \le 2 + 4 = 6 \quad \Longrightarrow \quad z(20) \le 6. \end{aligned}$$

2 Elementary invariances

Lemma 2.1 (Induced-subgraph monotonicity)**.** *If H is an induced subgraph of G, then $\zeta(H) \le \zeta(G)$.*

Proof. Restrict a cocolouring of G to $V(H)$. Every resulting class is a subset of a clique or of an independent set, hence is still homogeneous. Empty classes may be discarded. □

Lemma 2.2 (Complement invariance)**.** *A vertex set is homogeneous in G if and only if it is homogeneous in $\overline{G}$. Consequently, the property*

"has two vertex-disjoint homogeneous 4-sets"

is invariant under complementation.

Proof. Complementation exchanges cliques and independent sets and leaves the underlying vertex sets unchanged. □

3 The lower bound: a hand proof

Let

$$Q = \{1, 2, 4, 8, 9, 13, 15, 16\} \subset \mathbb{Z}_{17}$$

be the set of nonzero quadratic residues. The Paley graph P_{17} has vertex set $\mathbb{Z}_{17}$, with $i \sim j$ precisely when $i - j \in Q$. Since $-1 = 16 \in Q$, this defines an undirected graph.

Lemma 3.1. *The Paley graph P_{17} contains neither a K_4 nor an independent 4-set.*

Proof. Translations and multiplications by a nonzero square are automorphisms of P_{17}. They act transitively on edges: after translating one endpoint to 0, multiplication by the inverse of the edge difference sends the other endpoint to 1. Hence a hypothetical K_4 may be assumed to contain the edge $\{0, 1\}$.

The common neighbours of 0 and 1 are

$$Q \cap (1 + Q) = \{2, 9, 16\}.$$

Their pairwise differences are congruent, up to sign, to 7 or 14; neither lies in Q. Thus the three common neighbours are pairwise nonadjacent, so no two of them can complete $\{0, 1\}$ to a K_4.

Multiplication by any quadratic nonresidue, for example 3, sends residues to nonresidues and is an isomorphism $P_{17} \cong \overline{P_{17}}$. The absence of a K_4 therefore also implies the absence of an independent 4-set. □

Proposition 3.2. $z(20) \geq 6$.

Proof. By lemma 3.1, every homogeneous set in P_{17} has at most three vertices. Five homogeneous classes could therefore cover at most 15 vertices, so

$$\zeta(P_{17}) \geq \left\lceil \frac{17}{3} \right\rceil = 6.$$

Let $G = P_{17} \sqcup 3K_1$. The graph G has 20 vertices and contains P_{17} as an induced subgraph. By lemma 2.1, $\zeta(G) \geq 6$, proving $z(20) \geq 6$. □

For completeness, the following partition is an explicit six-class cocolouring of G, where $17, 18, 19$ are the isolated vertices:

$$\begin{aligned} I_1 &= \{0, 3, 6, 17, 18, 19\}, & I_2 &= \{1, 4, 7\}, & I_3 &= \{2, 9, 12\}, \\ I_4 &= \{5, 8, 11\}, & I_5 &= \{10, 13, 16\}, & K_1 &= \{14, 15\}. \end{aligned}$$

The I_i are independent sets and K_1 is an edge. No solver is needed for the lower bound.

4 Two disjoint homogeneous 4-sets

The classical identity $R(4,4) = 18$ is due to Greenwood and Gleason [3]; it is also recomputed by the catalogue run described in section 6.

Lemma 4.1. *Every graph on 20 vertices contains two vertex-disjoint homogeneous 4-sets.*

Proof. Suppose for contradiction that a 20-vertex graph G has no two vertex-disjoint homogeneous 4-sets. Since $20 \geq R(4,4) = 18$, it contains a homogeneous 4-set A. By lemma 2.2, complementing G if needed does not affect the target property, so assume that A is a clique. Relabel $A = \{a_0, a_1, a_2, a_3\}$ and put $B = V(G) \setminus A$.

If $G[B]$ contains a homogeneous 4-set, it is disjoint from A and the conclusion follows. Otherwise $G[B]$ is a $(4,4)$-graph on 16 vertices. The complete catalogue in section 6 gives, up to isomorphism, exactly the following two cores in graph6 notation:

$$C_0 = \texttt{OsHHirKdlp[IFVI|KpqfR},$$
$$C_1 = \texttt{Ov?IXZIhlRWjUXL[iphst}.$$

Each has 60 edges, degree sequence $7^8 8^8$, 56 triangles, and 56 independent triples.

After fixing $G[B]$ to C_0 or C_1, the only undetermined edges are the $4 \cdot 16 = 64$ edges between A and B. For each core, section 5 constructs a CNF whose satisfying assignments are *exactly* those cross-edge patterns for which the resulting graph has no two disjoint homogeneous 4-sets. The actual cross-edge pattern of G would therefore satisfy one of these CNFs, but both are unsatisfiable by the certificates in section 7. This contradiction proves the claim.

Notice that A itself is a homogeneous 4-set. It need not be encoded. The set B contains no homogeneous 4-set, so A cannot be one half of a disjoint pair. Every other possible pair consists of two mixed sets, which the CNF treats exhaustively. □

5 The exact fixed-core encoding

Fix $A = \{a_0, a_1, a_2, a_3\}$ as a clique and let $B = \{b_0, \ldots, b_{15}\}$ induce one of the two fixed cores. For $0 \leq i < 4$ and $0 \leq j < 16$, introduce

$$x_{i,j} = \begin{cases} 1, & a_i b_j \in E(G), \\ 0, & a_i b_j \notin E(G). \end{cases}$$

Lemma 5.1 (Mixed-set classification)**.** *The mixed 4-sets that can be homogeneous are exactly the four types in table 1.*

Proof. If a 4-set contains three vertices of A, those vertices already form a triangle; the set can only be a clique, and this occurs precisely when the three cross edges are present. If it contains two vertices of A, their edge rules out independence. The set is a clique precisely when the B-pair is an edge and the four cross edges are present.

With one vertex of A, the B-triple must be either a triangle, followed by three present cross edges, or an independent triple, followed by three absent cross edges. A 4-set wholly in B is not homogeneous by the definition of the core. The only 4-set wholly in A is A itself, already handled in the last paragraph of lemma 4.1. This proves both necessity and sufficiency. □

Table 1: Conditions under which a mixed 4-set is homogeneous.

Split	Necessary and sufficient condition	Count
$3A + 1B$	Clique: the three relevant cross edges are present.	$\binom{4}{3} \cdot 16 = 64$
$2A + 2B$	Clique: the B-pair is one of the 60 core edges and all four relevant cross edges are present.	$\binom{4}{2} \cdot 60 = 360$
$1A + 3B$	Clique: the B-triple is one of the 56 triangles and all three cross edges are present.	$4 \cdot 56 = 224$
$1A + 3B$	Independent: the B-triple is one of the 56 independent triples and all three cross edges are absent.	$4 \cdot 56 = 224$
	Total	**872**

Represent each possible mixed homogeneous set S by the conjunction C_S of its three or four required cross-edge literals. Whenever $S \cap T = \varnothing$, add the clause

$$\neg C_S \ \vee \ \neg C_T.$$

Since C_S and C_T are conjunctions, this is a single clause containing the negations of all their literals. For example, if

$$S = \{a_0, a_1, a_2, b_5\} \quad \text{and} \quad T = \{a_3, b_1, b_4, b_9\},$$

where $\{b_1, b_4, b_9\}$ is a core independent triple, the clause is

$$\neg x_{0,5} \vee \neg x_{1,5} \vee \neg x_{2,5} \vee x_{3,1} \vee x_{3,4} \vee x_{3,9}.$$

Proposition 5.2 (Encoding equivalence)**.** *For either core, an assignment to the* 64 *variables satisfies the constructed CNF if and only if the corresponding* 20*-vertex graph has no two vertex-disjoint homogeneous* 4*-sets.*

Proof. By lemma 5.1, every candidate other than A is represented by exactly one condition C_S, and the condition holds exactly when S is homogeneous. The clause for a disjoint pair S, T is false exactly when both C_S and C_T hold. Conjoining these clauses over all disjoint pairs therefore forbids exactly, and only, the existence of two disjoint homogeneous 4-sets. □

For each core, the resulting DIMACS formula has 64 variables and 104,524 clauses. The clause-length histogram is

$$43{,}840 \text{ of length } 6, \qquad 52{,}416 \text{ of length } 7, \qquad 8{,}268 \text{ of length } 8.$$

The only possible lengths are $6, 7, 8$, the sums of two three- or four-literal conditions subject to disjointness.

6 Catalogue generation

McKay's maintained Ramsey catalogue lists the number of non-isomorphic $(4, 4)$-graphs at each order and gives two at order 16 [4]. The local proof does not rely only on downloading those two graphs: `nauty/gen44.c` regenerates the sequence from the definition, using nauty for canonical labelling [5].

6.1 Why the generation is exhaustive

Suppose a complete set of isomorphism-class representatives is known at order n. Every $(4,4)$-graph H on $n+1$ vertices becomes a $(4,4)$-graph after deleting any vertex v. Hence H arises from some order-n representative by adjoining one vertex with some neighbourhood S. Trying every admissible S, canonicalising the result, and retaining one copy of each canonical form therefore reaches every order-$n+1$ isomorphism class. Induction starts with the unique one-vertex graph.

For a new vertex with neighbourhood S, a new homogeneous 4-set must contain that vertex. Admissibility is therefore equivalent to

$$G[S] \text{ contains no triangle} \quad \text{and} \quad G[V \setminus S] \text{ contains no independent triple.}$$

The implementation prunes to $n - 8 \le |S| \le 8$. Here is a short proof of the needed bound, included because it is otherwise a hidden $R(3,4) \le 9$ input. In a triangle-free graph on nine vertices with independence number at most three, every degree is at most three, since the neighbours of any vertex are independent. If some vertex v had degree at most two, deleting v and its neighbours would leave at least six vertices. The elementary $R(3,3) \le 6$ argument gives an independent triple among those remaining vertices, and adjoining v would give an independent 4-set. Thus every degree would have to be three, impossible on nine vertices by the handshaking lemma. Therefore a triangle-free set with independence number at most three has size at most eight. For completeness, the six-vertex argument is this: at any vertex of a red–blue K_6, three incident edges share a colour; their other endpoints either contain an edge of that colour or form a triangle of the other colour. Thus $R(3,3) \le 6$.

For $V \setminus S$, the second admissibility condition says that $\overline{G[V \setminus S]}$ is triangle-free, while the absence of a K_4 in the core says that this complement has independence number at most three. The same bound gives $|V \setminus S| \le 8$, hence $|S| \ge n - 8$.

6.2 Replayed output

The full computation was replayed on 26 July 2026. The counts, always up to isomorphism, were:

n	1	2	3	4	5	6	7	8	9
#	1	2	4	9	24	84	362	2,079	14,701

n	10	11	12	13	14	15	16	17	18
#	103,706	546,356	1,449,166	1,184,231	130,816	640	2	1	0

The two order-16 graph6 strings are exactly C_0, C_1 above. The order-17 survivor supplies $R(4,4) > 17$, while the empty order-18 level supplies $R(4,4) \le 18$; together they re-establish $R(4,4) = 18$.

7 Certificate verification

For each fixed core, a clause-learning-free DPLL program produced a tree-like DRUP proof. Core 0 visited 1,571 search nodes, made 785 decisions, and emitted 1,571 proof lines; core 1

visited 789 nodes, made 394 decisions, and emitted 789 proof lines. The preserved formulas have hashes

C_0 : `cbfea7b0b2cb712ad8b4f8b129f74c7f010e879064414eaef76e74aa88be3f44`,

C_1 : `d3d96a3c2a719c753242673b81aabc3977c5e435bb5996b4c58c85a82d11dc5a`.

The following checks were replayed on 26 July 2026.

Table 2: Replayed validation of the two fixed-core refutations.

Check	Core 0	Core 1
Separately implemented formula construction	872 conditions; 104,524 clauses; clause set identical	872 conditions; 104,524 clauses; clause set identical
Direct semantic spot-check	300/300 assignments agreed	300/300 assignments agreed
Separately implemented RUP checker	1,571 steps; empty clause derived	789 steps; empty clause derived
Package RUP checker	`VERIFIED`	`VERIFIED`
`drat-trim` [6]	`s VERIFIED`; 0 RAT lemmas	`s VERIFIED`; 0 RAT lemmas
`cake_lpr` [7]	`s VERIFIED UNSAT`	`s VERIFIED UNSAT`

The separately regenerated formulas are byte-identical to the certificate-package formulas. The package audit additionally tested 1,000 deterministic random assignments per core. The package's twelve-entry SHA-256 manifest passes in full.

The DRUP proofs were converted to LRAT with `drat-trim -L` and the resulting LRAT proofs were checked against the original CNFs by `cake_lpr`. Once the LRAT proof is replayed against the CNF, the translator need not be trusted for logical soundness: a faulty translation would fail unless it happened to output another valid proof.

As a negative control, cross-pairing core 0's formula with core 1's proof was rejected by the standalone RUP checker, by `drat-trim`, and by `cake_lpr`. One operational detail matters: `cake_lpr` can print a failure while returning process status 0. Automation must require the exact success verdict `s VERIFIED UNSAT`, not merely a zero exit status.

8 The residual 12-vertex graph

Lemma 8.1. $z(8) \leq 3$.

Computational proof. The program `geng` generated one representative of each of the 12,346 isomorphism classes of simple graphs on eight vertices. The exact recursive program `nauty/cochk.c` found every representative 3-cocolourable. □

Lemma 8.2. $z(12) \leq 4$.

Proof. Let G have 12 vertices. If G contains a homogeneous 4-set, use it as one class; by lemma 8.1, the remaining eight vertices require at most three classes.

Otherwise G is a $(4,4)$-graph. The orderly generator produced one representative of every one of the 1,449,166 isomorphism classes of such graphs on 12 vertices, and `nauty/cochk.c` found each representative 4-cocolourable. These two cases are exhaustive. □

The recursive decision procedure is exact at the algorithmic level: choose the least remaining vertex v, enumerate every clique and every independent set containing v, remove the chosen block, and recurse with one fewer available class. Every cocolouring has exactly one class containing v, so no partition is omitted. The assurance caveat is implementation correctness, not an intentionally incomplete search.

9 Completion of the upper bound

Proposition 9.1. $z(20) \leq 6$.

Proof. Let G have 20 vertices. By lemma 4.1, choose two disjoint homogeneous 4-sets and use them as two cocolouring classes. The remaining 12 vertices induce a graph cocolourable in at most four classes by lemma 8.2. Thus $\zeta(G) \leq 2+4 = 6$. Since G was arbitrary, $z(20) \leq 6$. □

Proof of theorem 1.2. Proposition 3.2 gives $z(20) \geq 6$, while proposition 9.1 gives $z(20) \leq 6$. □

$$\boxed{z(20) = 6}$$

10 Assurance boundary

This section is part of the proof report, not a disclaimer to be read after the fact. Different components carry different assurance.

Layer	What is established, and what remains trusted
Hand mathematics	The invariances, Paley lower bound, catalogue-completeness induction, mixed-set classification, CNF equivalence, and final deductions can be checked directly from this text.
SAT proof objects	Each DRUP refutation is accepted by two separately implemented RUP checkers and by `drat-trim`; the corresponding LRAT proof is accepted against the exact CNF by the formally verified `cake_lpr` core. This establishes unsatisfiability of those CNFs.
Semantic bridge	The graph-to-CNF equivalence is proved in proposition 5.2 and is also tested by a separately written direct graph implementation. The source implementation of that equivalence is not formally verified.
Catalogue	The generation is exhaustive if its implementation and nauty canonicalisation are correct. The complete sequence was replayed and matches McKay's maintained data, but the C program and nauty are not proved correct inside a proof assistant.
Cocolourability	The recursive algorithm is exhaustive, and the full 8- and 12-vertex runs were replayed. The C implementation is not formally verified.
System	The compiler, operating system, file I/O, and hardware remain in the conventional trusted computing base, except where the CakeML checker reduces that base for the SAT-checking layer.

Two artifact-specific caveats should be visible to reproducers.

- The recorded SHA-256 of `cake_lpr_arm8.S` matches the preserved assembly source. The current `basis_ffi.c` hash does not match the older recorded hash. The repository's explanation that this is a stale manifest for a small I/O shim is plausible, but is not established solely by the local files.
- `verify_certificate.py` contains a hard-coded temporary source-package path. That directory existed at replay time, and its two CNFs and two DRUP files were byte-identical to the durable package copies. The script completed successfully, but is not portable as written.

There is no Lean theorem of $z(20) = 6$ in the bundle. `lean_blueprint.md` is a design document that explicitly leaves the catalogue, residual enumeration, and graph-to-CNF reflection as formalisation obligations.

> **Status.** The accurate present claim is: *a complete, locally regenerated candidate computer-assisted proof of $z(20) = 6$, with certificate-backed SAT refutations and an explicit trusted-computing boundary.*

11 Reproduction record and artifact map

11.1 Versions bound to this note

- Local bundle: `/Users/admin/z20_repro`, Git commit `61e8300`.
- Public artifact snapshot: https://github.com/ipitchford/z20-cochromatic, commit `8d9829edebecdfc78384eef5b41617ddd32b11ea`.

- Verification date: 26 July 2026, macOS on arm64.
- Companion replay receipt: `z20_equals_6_replay_receipt_2026-07-26.md`.

11.2 Load-bearing local files

Relative path under `z20_repro`	Role
`nauty/gen44.c`	Orderly generator for all $(4,4)$-graphs, using nauty canonical forms.
`nauty/gen44d.c`	Dumping variant used to stream every order-12 representative.
`nauty/cochk.c`	Exact recursive k-cocolourability decision procedure.
`certificate/z20_core_certificate_candidate/generate.py`	Deterministic fixed-core CNF generator.
`certificate/z20_core_certificate_candidate/z20_core0.cnf` and `certificate/z20_core_certificate_candidate/z20_core1.cnf`	The two 64-variable formulas.
`certificate/z20_core_certificate_candidate/z20_core0.drup` and `certificate/z20_core_certificate_candidate/z20_core1.drup`	Tree-like RUP/DRUP refutations.
`verify_certificate.py`	Separately implemented formula construction, semantic comparison, and RUP checker; path repair is required as noted above.
`drat/drat-trim.c`	Source of the standard DRAT checker and LRAT converter.
`drat/mine_core0.cnf` and `drat/mine_core1.cnf`	Independently regenerated, byte-identical CNFs.
`cake_lpr/cake_lpr_arm8.S`	Preserved CakeML-compiled checker assembly.
`cake_lpr/core0.lrat` and `cake_lpr/core1.lrat`	LRAT proofs accepted by `cake_lpr`.
`nauty/gen44_result.txt`, `nauty/z12_result.txt`, and `nauty/lower_bound_result.txt`	Stored computation transcripts.

11.3 Minimal verdict checks

The certificate package documents exact build commands. A reproducer should, at minimum, check all of the following:

1. the package SHA-256 manifest;
2. exact equality of regenerated and packaged CNFs;
3. the package RUP checker on both formula/proof pairs;
4. `drat-trim` reporting `s VERIFIED` on both;

5. `cake_lpr` printing the exact verdict `s VERIFIED UNSAT` on both;
6. a cross-core negative control that is rejected;
7. the catalogue counts and exact order-16 graph6 strings;
8. the 12,346-representative $z(8)$ run and the 1,449,166-representative residual $z(12)$ run.

12 Provenance and AI disclosure

The local repository reports that every mathematical step in the result was produced by AI systems, while the human participant selected the problem and mediated between systems. The reported division of labour is: an unsuccessful Hermes-driven CEGIS search using GPT-5.6 Sol; subsequent instrumentation and survivor analysis by Claude 4.8; the two-core reduction, CNF generator, DPLL solver, and DRUP certificates from GPT-5.6 Sol; and verification by a different AI system, catalogue and residual enumeration, lower-bound reconstruction, and the earlier write-ups by Claude Opus 5. This paragraph records repository provenance; it is not an externally audited authorship determination.

This PDF was prepared with OpenAI Codex from the local proof manuscript, walkthrough, source code, certificates, logs, and fresh replays listed above. Codex revised the exposition, corrected assurance wording, added the hidden catalogue-pruning lemma, and typeset the document. The status label on the title page is intentionally retained. Any claim that the problem is settled, or any stronger publication claim, should await independent human review.

References

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